

AI FAQ ASSISTANT API

Data Flow Diagram

Request and data movement

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TEAM SIZE	5
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Context

Client/Postman -> Express API. MongoDB stores User and FAQ documents. Protected routes verify JWT. AI service sends question/topic prompts to Google Gemini and returns generated content.

Authentication

Register input -> validation -> duplicate email lookup -> User save with bcrypt password hook -> response with JWT. Login -> find user -> compare password -> response token. Protected profile loads the authenticated user.

FAQ flow

Public GET/list/search -> FAQ query -> populated creator name/email -> JSON response. Protected create sets createdBy to current user; update/delete first check FAQ creator ownership.

AI flow

JWT check -> body validation -> Gemini service generates text or structured JSON -> controller responds with success and output.

Configuration and privacy

MongoDB URI, JWT secret, Gemini key, and port come from environment settings. Do not distribute real .env credentials; source .env is excluded from the docs bundle.